Time Series — Solutions

Practice Exam — Week 12: Non-Linear Time Series — ARCH & GARCH

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Problem 1 (5 points)

Statement.

(ARCH(2): conditional vs. unconditional moments and stationarity.) *Let $\{r_t\}$ follow an* ARCH(2) *model*

$$r_t = \sigma_t \eta_t, \qquad \sigma_t^2 = \beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2,$$

with $\beta_0 > 0,\ \beta_1, \beta_2 \geq 0$, where $\{\eta_t\}$ is a white noise of mean 0 *and* unit *variance, independent of $\mathcal{F}_{t-1}$ (the information up to time $t-1$). Here σ_t is $\mathcal{F}_{t-1}$-measurable. Take the numerical values $\beta_0 = 0.2$, $\beta_1 = 0.3$, $\beta_2 = 0.5$.*

(a) *Using the law of iterated expectation (condition on $\mathcal{F}_{t-1}$), show that $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0$ and deduce $\mathbb{E}[r_t] = 0$. State also the conditional variance $\mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$.* *(1.5 pts)*

(b) *Assuming $\{r_t\}$ is (weakly) stationary, use the law of total variance to derive the marginal variance $\mathrm{Var}(r_t)$ as a function of $\beta_0, \beta_1, \beta_2$, naming every probability law you use. Then give its numerical value for the parameters above.* *(2 pts)*

(c) *State the necessary condition on β_1, β_2 for a (positive, finite) stationary variance to exist, check it for the given numbers, and explain in one sentence why this is the $p = 2$, $q = 0$ instance of the general $\sum_j \alpha_j + \sum_j \beta_j < 1$ rule.* *(1.5 pts)*

Solution.

(a) Since σ_t is $\mathcal{F}_{t-1}$-measurable and η_t is independent of $\mathcal{F}_{t-1}$ with $\mathbb{E}[\eta_t] = 0$,

$$\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[\sigma_t \eta_t \mid \mathcal{F}_{t-1}] = \sigma_t\, \mathbb{E}[\eta_t] = \sigma_t \cdot 0 = 0.$$

By the law of iterated (tower) expectation,

$$\mathbb{E}[r_t] = \mathbb{E}\big[\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]\big] = \mathbb{E}[0] = 0.$$

The conditional variance is, since $\mathbb{E}[r_t \mid \mathcal{F}_{t-1}] = 0$ and $\mathrm{Var}(\eta_t) = 1$,

$$\mathrm{Var}(r_t \mid \mathcal{F}_{t-1}) = \mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2\, \mathbb{E}[\eta_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2 = \beta_0 + \beta_1 r_{t-1}^2 + \beta_2 r_{t-2}^2.$$

(b) By the *law of total variance*,

$$\mathrm{Var}(r_t) = \mathbb{E}\big[\mathrm{Var}(r_t \mid \mathcal{F}_{t-1})\big] + \mathrm{Var}\big(\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]\big) = \mathbb{E}[\sigma_t^2] + \mathrm{Var}(0) = \mathbb{E}[\sigma_t^2].$$

Taking the (unconditional) expectation of the variance recursion and using *linearity of expectation*,

$$\mathbb{E}[\sigma_t^2] = \beta_0 + \beta_1 \mathbb{E}[r_{t-1}^2] + \beta_2 \mathbb{E}[r_{t-2}^2].$$

Because $\mathbb{E}[r_{t-j}] = 0$ we have $\mathbb{E}[r_{t-j}^2] = \mathrm{Var}(r_{t-j})$, and by *stationarity* $\mathrm{Var}(r_t) = \mathrm{Var}(r_{t-1}) = \mathrm{Var}(r_{t-2}) =: v$. Hence

$$v = \beta_0 + \beta_1 v + \beta_2 v \implies (1 - \beta_1 - \beta_2)\, v = \beta_0 \implies \boxed{\mathrm{Var}(r_t) = \frac{\beta_0}{1 - \beta_1 - \beta_2}} \quad (\beta_1 + \beta_2 < 1).$$

Numerically, with $\beta_0 = 0.2,\ \beta_1 = 0.3,\ \beta_2 = 0.5$:

$$\mathrm{Var}(r_t) = \frac{0.2}{1 - 0.3 - 0.5} = \frac{0.2}{0.2} = 1.$$

(c) For the variance to be positive and finite we need the denominator positive, i.e.

$$\boxed{\beta_1 + \beta_2 < 1} \qquad (\text{with } \beta_0 > 0,\ \beta_1, \beta_2 \geq 0).$$

Here $\beta_1 + \beta_2 = 0.3 + 0.5 = 0.8 < 1$, so a stationary variance exists (and equals 1). This is exactly the $p = 2$, $q = 0$ case of the general GARCH stationarity rule $\sum_j \alpha_j + \sum_j \beta_j < 1$: an ARCH(2) has α-coefficients β_1, β_2 and no GARCH (β_j) feedback terms, so the rule reads $\beta_1 + \beta_2 < 1$.

Problem 2 (5 points)

Statement.

(GARCH is mean-zero white noise; stationarity & long-run variance.) *A* GARCH(m, r) *process is*

$$r_t = \sigma_t\, \eta_t, \qquad \sigma_t^2 = \alpha_0 + \sum_{j=1}^{m} \alpha_j r_{t-j}^2 + \sum_{j=1}^{r} \beta_j \sigma_{t-j}^2,$$

with $\alpha_0 > 0$, $\alpha_j, \beta_j \geq 0$, and $\{\eta_t\}$ white noise of mean 0 and unit variance, independent of $\mathcal{F}_{t-1}$; σ_t is $\mathcal{F}_{t-1}$-measurable.

(a) *Prove that $\mathbb{E}[r_t] = 0$ and that, for every lag $\tau \neq 0$, $\mathrm{Cov}(r_{t+\tau}, r_t) = 0$, hence $\mathrm{Corr}(r_{t+\tau}, r_t) = 0$. (*Hint: *for $\tau > 0$ condition on $\mathcal{F}_{t+\tau-1}$ and pull out the measurable factors $r_t, \sigma_{t+\tau}$.) Conclude that $\{r_t\}$ is white noise.* *(2.5 pts)*

(b) *For a stationary GARCH, derive the unconditional variance $\mathrm{Var}(r_t) = \alpha_0 / \big(1 - \sum_j \alpha_j - \sum_j \beta_j\big)$ by taking unconditional expectations of the σ_t^2 recursion. State the resulting weak-stationarity condition. (*You may use $\mathbb{E}[r_{t-j}^2] = \mathbb{E}[\sigma_{t-j}^2] = \mathrm{Var}(r_t)$ at stationarity; justify this identity briefly.*)* *(1.5 pts)*

(c) *A* GARCH(1,1) *is fitted to daily log-returns with $\alpha_0 = 2 \times 10^{-5}$, $\alpha_1 = 0.10$, $\beta_1 = 0.85$. Verify weak stationarity, compute the unconditional variance and the implied daily volatility $\sqrt{\mathrm{Var}(r_t)}$, and comment in one sentence on the persistence $\alpha_1 + \beta_1$.* *(1 pt)*

Solution.

(a) *Zero mean.* σ_t is determined by the past, hence $\mathcal{F}_{t-1}$-measurable, and η_t is independent of $\mathcal{F}_{t-1}$ with $\mathbb{E}[\eta_t] = 0$. By the tower property,

$$\mathbb{E}[r_t] = \mathbb{E}\big[\mathbb{E}[r_t \mid \mathcal{F}_{t-1}]\big] = \mathbb{E}\big[\sigma_t\, \mathbb{E}[\eta_t \mid \mathcal{F}_{t-1}]\big] = \mathbb{E}[\sigma_t \cdot 0] = 0.$$

Uncorrelated. Fix $\tau > 0$. Since $\mathbb{E}[r_t] = 0$, $\mathrm{Cov}(r_{t+\tau}, r_t) = \mathbb{E}[r_{t+\tau} r_t]$. Both r_t and $\sigma_{t+\tau}$ are $\mathcal{F}_{t+\tau-1}$-measurable (they depend only on information up to time $t + \tau - 1$), so conditioning on $\mathcal{F}_{t+\tau-1}$ and pulling them out of the inner expectation,

$$\begin{aligned}\mathbb{E}[r_{t+\tau} r_t] &= \mathbb{E}\big[\mathbb{E}[\sigma_{t+\tau}\eta_{t+\tau}\, r_t \mid \mathcal{F}_{t+\tau-1}]\big] \\ &= \mathbb{E}\big[r_t\, \sigma_{t+\tau}\, \mathbb{E}[\eta_{t+\tau} \mid \mathcal{F}_{t+\tau-1}]\big] = \mathbb{E}[r_t\, \sigma_{t+\tau} \cdot 0] = 0,\end{aligned}$$

using $\mathbb{E}[\eta_{t+\tau} \mid \mathcal{F}_{t+\tau-1}] = \mathbb{E}[\eta_{t+\tau}] = 0$ by independence. By symmetry of the covariance the same holds for $\tau < 0$. Therefore $\mathrm{Cov}(r_{t+\tau}, r_t) = 0$ and $\mathrm{Corr}(r_{t+\tau}, r_t) = 0$ for all $\tau \neq 0$. Combined with $\mathbb{E}[r_t] = 0$ and a constant (stationary) variance, $\{r_t\}$ is **white noise**.

(b) As in Problem 1, $\mathbb{E}[r_{t-j}^2] = \mathrm{Var}(r_{t-j})$ (mean zero) and, using $\mathbb{E}[r_{t-j}^2 \mid \mathcal{F}_{t-j-1}] = \sigma_{t-j}^2$ together with the tower property, $\mathbb{E}[r_{t-j}^2] = \mathbb{E}[\sigma_{t-j}^2]$; by stationarity all these equal $v := \mathrm{Var}(r_t)$. Taking unconditional expectations of the recursion,

$$\mathbb{E}[\sigma_t^2] = \alpha_0 + \sum_{j=1}^{m} \alpha_j\, \mathbb{E}[r_{t-j}^2] + \sum_{j=1}^{r} \beta_j\, \mathbb{E}[\sigma_{t-j}^2] \implies v = \alpha_0 + \Big(\sum_j \alpha_j + \sum_j \beta_j\Big) v.$$

Solving,

$$\boxed{\mathrm{Var}(r_t) = v = \frac{\alpha_0}{1 - \sum_j \alpha_j - \sum_j \beta_j}},$$

which is positive and finite **iff**

$$\boxed{\sum_{j=1}^{m}\alpha_j+\sum_{j=1}^{r}\beta_j<1}\qquad\text{(the weak-stationarity condition).}$$

(c) For GARCH$(1,1)$ with $\alpha_0=2\times10^{-5},\ \alpha_1=0.10,\ \beta_1=0.85$:

$$\alpha_1+\beta_1=0.10+0.85=0.95<1,$$

so the process is weakly stationary. Then

$$\operatorname{Var}(r_t)=\frac{\alpha_0}{1-\alpha_1-\beta_1}=\frac{2\times10^{-5}}{1-0.95}=\frac{2\times10^{-5}}{0.05}=4\times10^{-4},$$

and the implied daily volatility is

$$\sqrt{\operatorname{Var}(r_t)}=\sqrt{4\times10^{-4}}=2\times10^{-2}=2\%\text{ per day},$$

a realistic equity figure. The persistence $\alpha_1+\beta_1=0.95$ is close to 1, the hallmark of financial GARCH fits: volatility shocks decay only slowly (sticky volatility, long quiet/turbulent regimes), yet the process stays stationary because the sum is still strictly below 1.

Problem 3 (5 points)

Statement.

(ARCH(1): kurtosis, the ACF of the squares, and a volatility forecast.) *Let $\{r_t\}$ be a stationary* ARCH(1),

$$r_t=\sigma_t\,\eta_t,\qquad \sigma_t^2=\alpha_0+\alpha_1 r_{t-1}^2,$$

with $\alpha_0>0$, $0<\alpha_1<1$, and Gaussian *innovations $\eta_t\sim\mathcal{N}(0,1)$ independent of $\mathcal{F}_{t-1}$. Recall $\mathbb{E}[r_t]=0$ and $\operatorname{Var}(r_t)=\alpha_0/(1-\alpha_1)$. You may use $\mathbb{E}[\eta_t^4]=3$.*

(a) *Prove that for $\tau>0$,*

$$\operatorname{Cov}\big(r_{t+\tau}^2,r_t^2\big)=\alpha_1\ \operatorname{Cov}\big(r_{t+\tau-1}^2,r_t^2\big),$$

by conditioning on $\mathcal{F}_{t+\tau-1}$ and using $\mathbb{E}[\eta^2\mid\mathcal{F}_{t+\tau-1}]=1$; be sure to show the constant terms cancel using $\mathbb{E}[r_t^2]=\alpha_0/(1-\alpha_1)$. Deduce that, provided $0<\alpha_1<1/\sqrt{3}$, $\operatorname{Corr}(r_{t+\tau}^2,r_t^2)=\alpha_1^{|\tau|}$. *(2.5 pts)*

(b) *The kurtosis of an* ARCH(1) *is $\kappa=\dfrac{3(1-\alpha_1^2)}{1-3\alpha_1^2}$ (valid when $\alpha_1^2<1/3$). An estimated model has $\alpha_1=0.5$. Compute κ, comment on the tails versus the normal, and report $\operatorname{Corr}(r_{t+\tau}^2,r_t^2)$ at lags $\tau=1,2,3$. Why does the constraint for a finite fourth moment ($\alpha_1^2<1/3$) differ from the one for a finite variance ($\alpha_1<1$)?* *(1.5 pts)*

(c) *For the same model take $\alpha_0=4\times10^{-4}$ and $\alpha_1=0.5$, and suppose the latest observed return is $r_t=0.04$. Give the one-step-ahead conditional variance σ_{t+1}^2 and a 95% predictive interval for r_{t+1} under Gaussian innovations; compare its width with the unconditional $\pm1.96\sqrt{\operatorname{Var}(r_t)}$ band and interpret.* *(1 pt)*

Solution.

(a) Fix $\tau>0$ and start from the covariance, using stationarity ($\mathbb{E}[r_{t+\tau}^2]=\mathbb{E}[r_t^2]$):

$$\operatorname{Cov}(r_{t+\tau}^2,r_t^2)=\mathbb{E}[r_{t+\tau}^2r_t^2]-\big(\mathbb{E}[r_t^2]\big)^2.$$

Condition on $\mathcal{F}_{t+\tau-1}$: both r_t^2 and $\sigma_{t+\tau}^2=\alpha_0+\alpha_1 r_{t+\tau-1}^2$ are measurable there, and $\mathbb{E}[\eta_{t+\tau}^2\mid\mathcal{F}_{t+\tau-1}]=\mathbb{E}[\eta_{t+\tau}^2]=1$, so

$$\mathbb{E}[r_{t+\tau}^2r_t^2\mid\mathcal{F}_{t+\tau-1}]=r_t^2\,\sigma_{t+\tau}^2\,\mathbb{E}[\eta_{t+\tau}^2\mid\mathcal{F}_{t+\tau-1}]=r_t^2\big(\alpha_0+\alpha_1 r_{t+\tau-1}^2\big).$$

Taking expectations,

$$\mathbb{E}[r_{t+\tau}^2 r_t^2] = \alpha_0\,\mathbb{E}[r_t^2] + \alpha_1\,\mathbb{E}[r_{t+\tau-1}^2 r_t^2].$$

Write $C_\tau := \mathrm{Cov}(r_{t+\tau}^2, r_t^2)$ and use $\mathbb{E}[r_{t+\tau-1}^2 r_t^2] = C_{\tau-1} + \big(\mathbb{E}[r_t^2]\big)^2$ (again by stationarity, $\mathbb{E}[r_{t+\tau-1}^2] = \mathbb{E}[r_t^2]$). Then

$$C_\tau + \big(\mathbb{E}[r_t^2]\big)^2 = \alpha_0\,\mathbb{E}[r_t^2] + \alpha_1\Big(C_{\tau-1} + \big(\mathbb{E}[r_t^2]\big)^2\Big),$$

i.e.

$$C_\tau = \alpha_1 C_{\tau-1} + \underbrace{\alpha_0\,\mathbb{E}[r_t^2] + \alpha_1\big(\mathbb{E}[r_t^2]\big)^2 - \big(\mathbb{E}[r_t^2]\big)^2}_{=:R}.$$

Substitute $\mathbb{E}[r_t^2] = \mathrm{Var}(r_t) = \alpha_0/(1-\alpha_1)$:

$$R = \alpha_0 \cdot \frac{\alpha_0}{1-\alpha_1} - (1-\alpha_1)\Big(\frac{\alpha_0}{1-\alpha_1}\Big)^2 = \frac{\alpha_0^2}{1-\alpha_1} - \frac{\alpha_0^2}{1-\alpha_1} = 0.$$

Hence

$$\boxed{C_\tau = \alpha_1\, C_{\tau-1}} \qquad (\tau > 0).$$

Iterating τ times from $C_0 = \mathrm{Var}(r_t^2)$ gives $C_\tau = \alpha_1^\tau\ \mathrm{Var}(r_t^2)$, so

$$\mathrm{Corr}(r_{t+\tau}^2, r_t^2) = \frac{C_\tau}{\mathrm{Var}(r_t^2)} = \alpha_1^\tau,$$

and by symmetry of the autocovariance, $\mathrm{Corr}(r_{t+\tau}^2, r_t^2) = \alpha_1^{|\tau|}$ for all τ. (The condition $0 < \alpha_1 < 1/\sqrt{3}$, i.e. $\alpha_1^2 < 1/3$, guarantees $\mathrm{Var}(r_t^2) = \mathbb{E}[r_t^4] - (\mathbb{E}[r_t^2])^2 < \infty$, so the correlation is well defined.)

(b) With $\alpha_1 = 0.5$, $\alpha_1^2 = 0.25 < 1/3$, so the fourth moment exists and

$$\kappa = \frac{3(1-\alpha_1^2)}{1-3\alpha_1^2} = \frac{3(1-0.25)}{1-0.75} = \frac{3\cdot 0.75}{0.25} = \frac{2.25}{0.25} = 9.$$

Since $\kappa = 9 > 3$, the marginal law has *heavier tails than the normal* (excess kurtosis $9-3=6$): ARCH generates fat tails endogenously, even with Gaussian innovations. The squared-return autocorrelations are $\mathrm{Corr}(r_{t+\tau}^2, r_t^2) = 0.5^{|\tau|}$:

$$\tau = 1:\ 0.5, \qquad \tau = 2:\ 0.25, \qquad \tau = 3:\ 0.125.$$

The two thresholds differ because they control different moments: $\alpha_1 < 1$ keeps the *second* moment ($\mathrm{Var}\, r_t = \alpha_0/(1-\alpha_1)$) finite, whereas the *fourth* moment involves $\mathbb{E}[\sigma_t^4]$ and the recursion for it carries the factor $3\alpha_1^2$ (from $\mathbb{E}[\eta^4] = 3$); a finite solution needs $3\alpha_1^2 < 1$, i.e. the strictly tighter $\alpha_1^2 < 1/3$. As $\alpha_1^2 \uparrow 1/3$ the kurtosis $\to \infty$.

(c) With $\alpha_0 = 4\times 10^{-4}$, $\alpha_1 = 0.5$ and $r_t = 0.04$ (so $r_t^2 = 1.6\times 10^{-3}$),

$$\sigma_{t+1}^2 = \alpha_0 + \alpha_1 r_t^2 = 4\times 10^{-4} + 0.5\,(1.6\times 10^{-3}) = 4\times 10^{-4} + 8\times 10^{-4} = 1.2\times 10^{-3},$$

so $\sigma_{t+1} = \sqrt{1.2\times 10^{-3}} \approx 0.0346$. Since $r_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}(0, \sigma_{t+1}^2)$, the 95% conditional predictive interval is

$$0 \pm 1.96\,\sigma_{t+1} = \pm 1.96\,(0.0346) \approx \pm 0.0679.$$

The unconditional variance is $\mathrm{Var}(r_t) = \alpha_0/(1-\alpha_1) = 4\times 10^{-4}/0.5 = 8\times 10^{-4}$, so $\sqrt{\mathrm{Var}(r_t)} \approx 0.0283$ and the unconditional band is $\pm 1.96\,(0.0283) \approx \pm 0.0554$. The conditional band ($\pm 0.068$) is markedly *wider* than the unconditional one (± 0.055): after the large move $r_t = 0.04$ the model raises its volatility forecast — exactly the volatility-clustering behaviour ARCH is designed to capture.

Problem 4 (5 points)

Statement.

(Revision of Week 11: diagnostics — Ljung–Box test and AIC/BIC selection.) *A mean-zero* ARMA$(1,1)$ *model is fitted to* $n = 100$ *observations. The residual sample autocorrelations at lags* 1–5 *are*

$$\hat{r}_1 = 0.05, \quad \hat{r}_2 = -0.12, \quad \hat{r}_3 = 0.08, \quad \hat{r}_4 = -0.10, \quad \hat{r}_5 = 0.06.$$

Recall the Ljung–Box statistic $Q_m = n(n+2)\sum_{\tau=1}^{m} \hat{r}_\tau^2/(n-\tau)$, *which under* $H_0 : \rho_1 = \cdots = \rho_m = 0$ *is approximately* χ^2_{m-p-q} *for an* ARMA(p,q) *fit. Useful quantiles:* $\chi^2_{2,0.95} = 5.991$, $\chi^2_{3,0.95} = 7.815$, $\chi^2_{4,0.95} = 9.488$.

(a) *Compute the Ljung–Box statistic* Q_5 *(show the per-lag terms).* *(1.5 pts)*

(b) *Test* $H_0 : \rho_1 = \cdots = \rho_5 = 0$ *at the* 5% *level: state the degrees of freedom, the critical value, the decision, and the practical conclusion about model adequacy. Would the decision change had an* AR(1) *been fitted instead?* *(1.5 pts)*

(c) *Recall* $\text{AIC} = -2\log L + 2k$ *and* $\text{BIC} = -2\log L + k\log n$ *with* $k = p+q+1$ *for an* ARMA(p,q). *A competing* AR(3) *(i.e.* $k = 4$*) is compared with the* ARMA$(1,1)$ *(i.e.* $k = 3$*) at* $n = 100$. *By how much must* $-2\log L$ *drop for AIC, respectively BIC, to prefer the larger model, and which criterion is more willing to select it? Why is in-sample* R^2 *a poor tool for this comparison?* *(2 pts)*

Solution.

(a) With $n = 100$, $n(n+2) = 100 \cdot 102 = 10200$, and $m = 5$:

$$Q_5 = 10200\left(\frac{0.05^2}{99} + \frac{(-0.12)^2}{98} + \frac{0.08^2}{97} + \frac{(-0.10)^2}{96} + \frac{0.06^2}{95}\right).$$

Term by term,

$$\frac{0.0025}{99} = 2.525 \times 10^{-5}, \quad \frac{0.0144}{98} = 1.469 \times 10^{-4}, \quad \frac{0.0064}{97} = 6.598 \times 10^{-5},$$
$$\frac{0.0100}{96} = 1.042 \times 10^{-4}, \quad \frac{0.0036}{95} = 3.789 \times 10^{-5}.$$

Summing the five terms gives 3.802×10^{-4}, so

$$Q_5 \approx 10200 \times 3.802 \times 10^{-4} \approx \boxed{3.88}.$$

(b) For an ARMA$(1,1)$, $p = q = 1$, so $\text{df} = m - p - q = 5 - 1 - 1 = 3$. The 5% critical value is $\chi^2_{3,0.95} = 7.815$. Since $Q_5 \approx 3.88 < 7.815$, we **do not reject** H_0: there is no significant evidence of residual autocorrelation up to lag 5, so the ARMA$(1,1)$ appears *adequate*.

Had an AR(1) been fitted, $p = 1, q = 0$, so $\text{df} = 5 - 1 - 0 = 4$ and the critical value rises to $\chi^2_{4,0.95} = 9.488$. Since $3.88 < 9.488$, the decision is unchanged (do not reject). In general fewer fitted parameters leave more degrees of freedom and widen the acceptance region.

(c) The parameter counts are $k = p+q+1$: for ARMA$(1,1)$, $k = 3$; for AR(3), $k = 4$ (one extra parameter). With $n = 100$, $\log n = \log 100 \approx 4.605$. Comparing penalties when adding the one extra parameter:

$$\Delta(\text{penalty})_{\text{AIC}} = 2\,(k_{\text{big}} - k_{\text{small}}) = 2,$$
$$\Delta(\text{penalty})_{\text{BIC}} = (\log n)(k_{\text{big}} - k_{\text{small}}) = \log 100 \approx 4.605.$$

So the AR(3) is preferred only if $-2\log L$ *drops* by more than the extra penalty:

AIC: drop > 2 (i.e. $\log L$ improves by > 1); BIC: drop > 4.605 ($\log L$ improves by > 2.30).

Since BIC's $\log n = 4.605 > 2$ charges more per parameter (as $n > e^2 \approx 7.4$), **AIC is more willing to select the larger** AR(3) **model**; BIC is harder to satisfy and favours the parsimonious ARMA$(1, 1)$.

In-sample $R^2 = 1 - s_e^2/s_y^2$ is a poor selection tool because adding parameters can only *increase* it (the larger nested model fits the training data at least as well), so a higher R^2 for AR(3) is no evidence of a genuinely better model; moreover even the true model has a bounded R^2 (a stochastic process is not perfectly predictable). Information criteria penalise complexity and are the correct basis for the comparison.